

end of its lifecycle.

Fourth is thermal requirement. You have to know your thermal requirements and the conditions in which the device will be used.

What kind of special processing capabilities or support is offered by Intel?

First of all, we have products which are specific to the embedded roadmap, not available on the PC space. For example, last year we introduced a version of Atom, Atom-XL, meant specifically for the portable embedded market.

Second, we have realised that we need to help the customer build the product. Let's take the case of portable devices. We study the market, pick up the end user requirements and work with a third party to build a motherboard suitable to end users' needs (like a small-form factor-motherboard for handheld devices). So now when we go to the customer, we tell them that we have the right silicon for them and we have enabled the motherboard. The only thing that the OEM needs to do is write the software and do the chassis. We help enable the most complicated part—the motherboard. We provide a 'reference design' for our target market segments, which is a document to explain the step-by-step process of developing the products and choosing the right components.

Third is our Intel India design house programme. This programme is meant for Indian companies, for designing in India. When customers sign up for this programme, they get a lot of technical support where our technical marketing engineers work closely with them to develop the entire product. They can enrol in an 'early access programme.' Under this programme, if the chip is coming out next year, we give them the plans and architecture specifications this year and train them in designing the motherboard. So they are ready to launch the product when the chip is announced.

How do you differentiate support to hobbyists, small companies, big firms and engineering students?

In India, there are three types of target audience for all our programmes. One is the Indian OEMs—people who are building products in India. In most cases, there is relatively little design work involved since customers purchase motherboards from one of the board vendors in our ecosystem. So it's more like a match-making process.

Second is the huge Indian design house community. These are companies which are actually designing the products. They need a lot of technical support in terms of reference designs, schematics and early action programmes. We provide bulk of our technical support in this area.

Lastly, we have hobbyists and the academic community. Currently, we have a programme called 'embedded Atom programme' only for India. Intel has an academic relation group which runs this programme. The idea is that the students and hobbyists can enter and make proposals.

We have a panel, and we meet to pick the top 25-30 ideas. We give development kits to the selected idea developers, where they can write the software. The whole idea is to actually build the entire product and showcase it.

'Connect this' is another worldwide community where embedded designers can share, vote for and discuss the ideas. The best idea is selected by a panel of judges and the community is awarded.

What are the pros and cons of a two-chip solution, especially in terms of performance versus pricing?

In a two-chip solution, one chip houses the CPU and memory controller and the other chip is the system controller hub, which is basically for the graphics and inputs/outputs. Intel originally started with three chips, now we have two chips and next we will have a single-chip solution (system on a chip). In terms of performance, there are a kind of value segment, the mid-range segment and then the high-end segment in general. Prior to Atom, Intel played in the high-range segment, where you need very high performance. With introduction of Atom, we are looking into mid range where we have mid-level performance built specially for portable devices.

Atom is a two-chip solution. There are lots of single-chip solutions from Nvidia, Freescale etc having similar performance. So how does Atom chip stand apart?

Atom is the fastest CPU in the sub-3-watt category. There is a large market that needs performance, and a two-chip solution is not that much of real estate. We have customers which have built motherboards of the size of credit cards. So Atom is very, very tiny. It's the tiniest chip that Intel has ever built. The biggest advantage is that Intel Atom is PC-compatible. People can write the program on the PC and export it to a completely different device. Your engineering work is significantly reduced due to backward compatibility.

Intel has entered the embedded segment many a times and then exited. But now we see a renewed focus in this segment. Why so?

Intel started its embedded business almost 30 years ago with introduction of the first microcontroller. We have always focused on the embedded segment and actually continued to grow the business. It's just that when you think of Intel, you think of a PC company. But Intel is also one of the largest embedded companies in the world. If Intel's embedded business would have been a separate company altogether, we would have been amongst Fortune 1000 in terms of revenue.

A few things that have happened in the recent past have increased our focus a little bit. The biggest thing we have is the Atom processor. It has allowed us to get into some new segments which we didn't focus on before.